Emulating the year-100 state of an idealised ocean model from two vertical-mixing constants

10 November 2026

Abstract

We test whether the temperature field that an idealised ocean model reaches after 100 years can be predicted directly from the two constants (c_k, c_ϵ) of its turbulent-kinetic-energy mixing closure. From 100 runs on a 10×10 logarithmic grid, a small neural network that maps the constants to 20 principal components of the state predicts held-out states with a root-mean-square error of 0.017–0.038 K in interpolation and 0.049–0.114 K in extrapolation, against 0.16–0.48 K for the mean state. The response collapses onto one combination, $s \approx \log c_k - 1.08 \log c_\epsilon$, as the scaling invariance of the closure predicts. The network improves on a Gaussian process on s by 13–17 % in interpolation, a margin that is not statistically established. Every method fails at the warm edge of the design, where the states leave the span of the training set. Restarted from an emulated state, the model keeps most of the error of that state for ten years. Away from the warm edge the state is worth 92–99 of the 100 years of spin-up.

1 Introduction

Calibrating the parameters of an ocean model requires its response to them after a long spin-up, and each evaluation costs a full integration. An emulator that maps the parameters to the spun-up state would replace that integration, for calibration or as an initial condition. Emulators of high-dimensional model output are usually built on a principal-component basis with one Gaussian process per coefficient [9]. Here we evaluate that construction with a neural network in place of the Gaussian processes, on a deliberately small problem: two constants of the vertical-mixing closure of an idealised channel-and-basin configuration, whose constants have been tuned in the global model by Bayesian optimisation [11]. We ask three questions. How accurately is a held-out state predicted, inside and outside the sampled range? Does the network improve on simpler emulators? Is a predicted state a useful initial condition?

2 Model and data

Model. The runs use mini_veros, a JAX port of Veros [6, 7] by E. Meunier, in its ACC configuration: a spherical grid of 2° cells (30×42, about 42°S to 42°N), 15 levels over 2080 m, and a meridional wall from 20°S to the northern boundary, which leaves a re-entrant channel to the south and a closed basin to the north. The forcing is steady: a zonal wind stress of up to $0.1\,\mathrm{N\,m^{-2}}$ over the channel and $0.2\,\mathrm{N\,m^{-2}}$ north of 10°N, and a restoring of surface temperature over 30 days to a profile that falls from 15 °C in the tropics to 0 °C at both ends. Salinity is uniform, so at a given depth density depends on temperature only. Mesoscale eddies are parameterised by isoneutral and thickness diffusion [5] with an eddy-energy closure [3] held at its default constants.

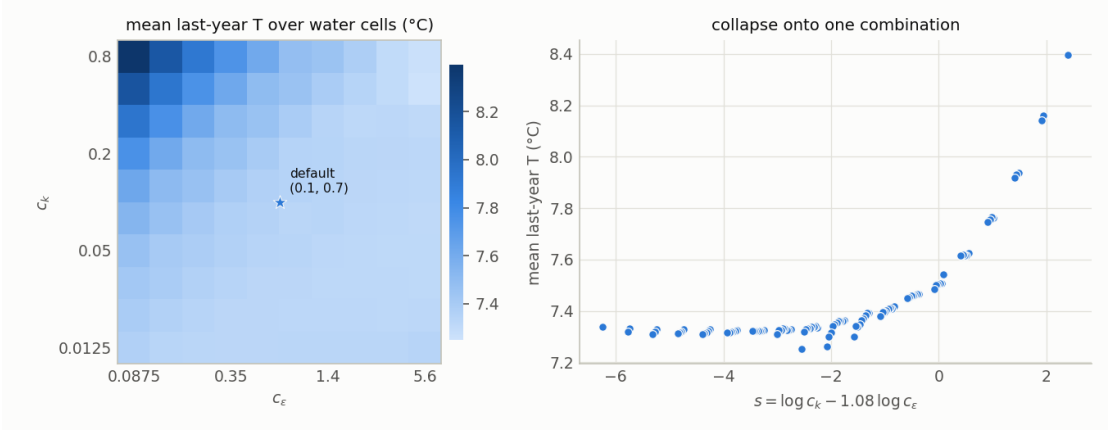


Figure 1: Domain-mean temperature of the target over the parameter grid (left; star: default constants) and against $s = \log c_k - 1.08 \log c_\epsilon$ (right).

Varied parameters. Vertical mixing follows the TKE closure of [4]. With e the turbulent kinetic energy and l the mixing length,

$$K_M = c_k \, l \sqrt{e}, \qquad \epsilon = c_\epsilon \, e^{3/2}/l, \qquad l = \sqrt{2e}/N, \tag{1}$$

where K_M is the vertical viscosity, ϵ the dissipation and N the buoyancy frequency [1]. Substituting l gives $K_M = \sqrt{2}\, c_k e/N$ and $\epsilon = c_\epsilon N e/\sqrt{2}$, which are unchanged under $(c_k, c_\epsilon, e) \to (\lambda c_k, \lambda c_\epsilon, e/\lambda)$. A steady state therefore depends on the ratio c_k/c_ϵ alone, except through the two terms that break this invariance: the vertical diffusion of e and the geometric bounds on l. The stored runs agree: along the line $c_k/c_\epsilon = 1/7$, e falls 45-fold as c_k rises from 0.0125 to 0.504 while $c_k e$ stays within $1.7\text{–}1.9 \times 10^{-7}\ \mathrm{m^2 s^{-2}}$.

Ensemble and target. The ensemble has 100 runs on a 10×10 grid, logarithmic in both constants, with $c_k \in [0.0125, 0.8]$ and $c_\epsilon \in [0.0875, 5.6]$, centred on the default $(0.1, 0.7)$. Each run starts from the same stratified state at rest and lasts 100 years. The target y is the mean temperature of its last 12 monthly snapshots on the 17 970 water cells. The runs are not equilibrated: at year 100 the state still changes by about 0.01 K per year, so the target is the state of this protocol, not an equilibrium.

Structure of the response. The first principal component of the 100 states is linear in $s = \log c_k - 1.08 \log c_\epsilon$, close to the logarithm of the invariant ratio (Figure 1). The response has two regimes. For $s < -2$ (48 runs) the domain-mean temperature varies by 0.09 K; above, it rises by 1.1 K towards the corner of large c_k and small c_ϵ, hereafter the warm corner.

3 Emulator and baselines

The emulator predicts a state in the span of the n training states,

$$\hat{y}(z) = \mu + \sum_{k=1}^{K} a_k(z)\, v_k, \qquad K = \min(20,\, n-1), \tag{2}$$

where z is the standardised pair $(\log c_k, \log c_\epsilon)$, and μ and v_k are the mean and the leading principal directions of the training states. The coefficients $a(z)$ come from a multilayer perceptron $2 \to 64 \to 64 \to K$, fitted by L-BFGS on the squared error of the coefficients with an L_2 penalty α on the weights [12]. The prediction is the mean of five networks that differ in their initialisation. The

activation and α are chosen on the training runs of each split by repeated five-fold cross-validation with the one-standard-error rule [8] (Appendix A). A fit takes seconds on one CPU core.

Five baselines use the same training runs, without tuning: the mean state; the nearest training run in z; a thin-plate spline interpolating every cell; a Gaussian process (GP) per principal coefficient on z, with a Matérn-5/2 kernel and one length scale per input [13]; and the same GP on the single input $s = w^\top z$, with w fitted on the training runs, a one-dimensional active subspace [2].

Evaluation. Each method is trained on part of the grid and scored on the rest, in eight splits. Five test interpolation: ten scattered interior runs, a band of three rows of constant c_k, the central 5×5 and 7×7 blocks, and leave-one-out over the 100 runs. Three test extrapolation: the top row ($c_k = 0.8$), and the 51 or 75 outer runs when only the central block is used for training. The score of a run is the root-mean-square error (RMSE) over the water cells, and the score of a split is its mean over the held-out runs. For the network it is further averaged over four sets of initialisations.

4 Results

4.1 Accuracy

The network has the lowest mean error on the eight splits (Table 1): 0.017–0.038 K in interpolation and 0.049–0.114 K in extrapolation, a tenth to a half of the error of the mean state. The GP on s follows, ahead of the GP on z and the spline; in interpolation the network is 13–17 % below it. The two emulators that know s are thus the two best, and the gain of the network over the GP on s (0.0025–0.0075 K) is small beside the gain of either over the nearest run.

Errors follow the two regimes of the response (Figure 2, left). In leave-one-out the median error of the network is 0.012 K where the response is flat ($s < -2$) and 0.023 K on the warm side, where the nearest run degrades to 0.10 K. In interpolation the error is concentrated in the upper 500 m (0.014–0.055 K, against 0.003–0.016 K below 1000 m).

4.2 Extrapolation and the warm edge

Outside the training range of s the error of every method rises together (Figure 2, right), up to 1 K at the warm corner. The cause is the basis, not the regression. Projecting a held-out truth on the training principal directions already leaves 0.010–0.017 K in interpolation and 0.031–0.079 K in extrapolation; at the warm corner, three grid cells from the nearest training run, it leaves 0.85 K of the network's 1.06 K, and every method scores between 1.06 and 1.45 K. The limit is shared by all methods that predict in the span of the training states [14].

Figure 3 shows the three situations. Inside the sampled range the network reproduces the pattern and the amplitude of the anomaly (correlation 0.98, amplitude 1.05). One row beyond it, at the warm corner, it still does (0.99, 0.99; 0.16 K). Three cells beyond, it keeps a weak version of the pattern with a third of the amplitude and a cold bias of −0.68 K.

Extrapolated fields are also less physical. Among the held-out predictions of the outer-51 split, the largest density inversion between two vertically adjacent cells is 0.82 kg m^{-3}, against 0.03 kg m^{-3} in the data.[1] On that split the network also has the lowest mean error on three of four integral diagnostics (channel density contrast, density spread at 106 m, north–south density difference at 490 m); the GP on s is better on the volume-mean temperature.

[1] Evaluation with the metrics of the companion repository MiniVeros-Diffusion (branch `deterministic-evaluation`, `Reports/Evaluation`), which also holds the restart experiment of Section 4.4.

split	held out / train	network	range	median	GP on s	GP on z	RBF	nearest	mean state
scattered	10 / 90	**0.017**	0.0008	0.013	0.019	0.022	0.021	0.038	0.161
band (three rows)	30 / 70	**0.038**	0.0010	0.033	0.045	0.051	0.057	0.118	0.204
centre 5×5	25 / 75	**0.025**	0.0004	0.019	0.029	0.035	0.037	0.080	0.162
centre 7×7	49 / 51	**0.029**	0.0006	0.026	0.035	0.069	0.068	0.116	0.203
leave-one-out	100 / 99	**0.024**	0.0004	0.017	0.028	0.032	0.034	0.085	0.221
top row	10 / 90	**0.049**	0.0028	0.033	0.079	0.083	0.084	0.176	0.482
outer 51	51 / 49	**0.067**	0.0005	0.027	0.074	0.081	0.093	0.134	0.265
outer 75	75 / 25	**0.114**	0.0009	0.041	0.125	0.131	0.138	0.159	0.226

Table 1: RMSE (K) over the water cells, mean over the held-out runs. Network: mean over four sets of initialisations, the range of that mean across the sets, and the median over the runs. Bold: lowest mean.

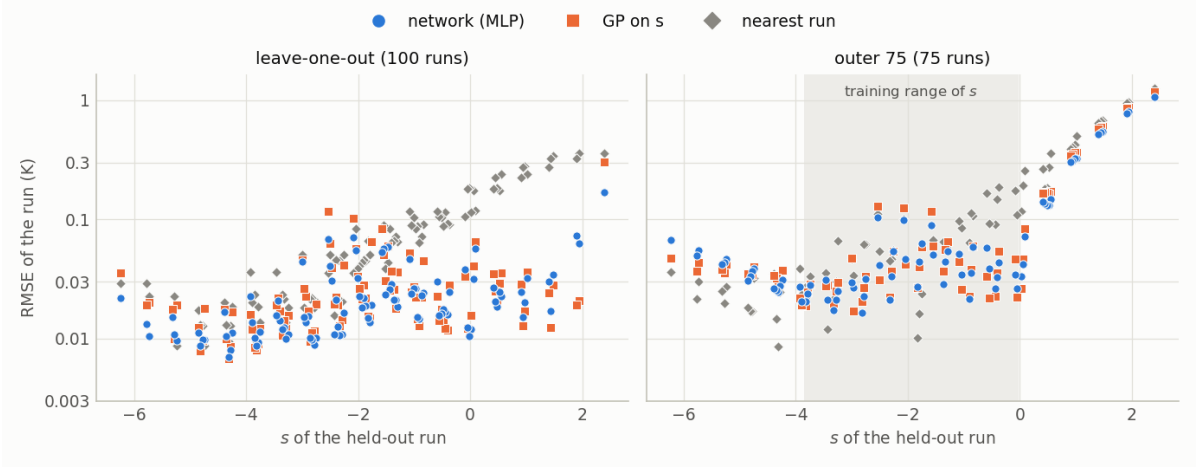


Figure 2: RMSE of each held-out run against its s. Left: leave-one-out. Right: the 75 outer runs, predicted from the central 5×5 block; the band is the range of s of the training runs.

4.3 How firm is the ranking?

The network and the GP on s are ahead of the other baselines on the eight splits, under both rules for choosing the hyper-parameters; the four sets of initialisations move a split mean by at most 0.003 K. The margin between the network and the GP on s is less firm, for three reasons.

First, it depends on the hyper-parameters. Configurations that cross-validation cannot separate differ by up to 0.03 K in test error. With the configuration of minimum cross-validation error instead of the one-standard-error rule, the network ties with the GP on s on the 7×7 block and on outer 75 (Appendix A).

Second, the held-out runs are not independent. A Wilcoxon test on the per-run differences, with a Holm correction over the eight splits [10], favours the network on six of them ($p \leq 0.044$). But runs on one diagonal of constant c_k/c_ϵ have nearly the same state, so the test overstates the evidence.

Third, three checks from an independent review, made with scripts that are not part of the repository, remain to be reproduced: with diagonals as the units of the test, no split is significant; linear interpolation of the fields along s is as accurate as the network in interpolation; and a GP on s and the direction orthogonal to it closes much of the remaining gap. Until they are, the supported conclusion is that the collapse onto s carries most of the skill, and that the network adds a consistent but unproven 13–17 %.

Two further limits concern use. The spread of the five networks (0.003–0.011 K) is far below the error and is not an uncertainty. With only four training runs cross-validation cannot choose the hyper-parameters, and the GP on s is then the safer emulator (Appendix B).

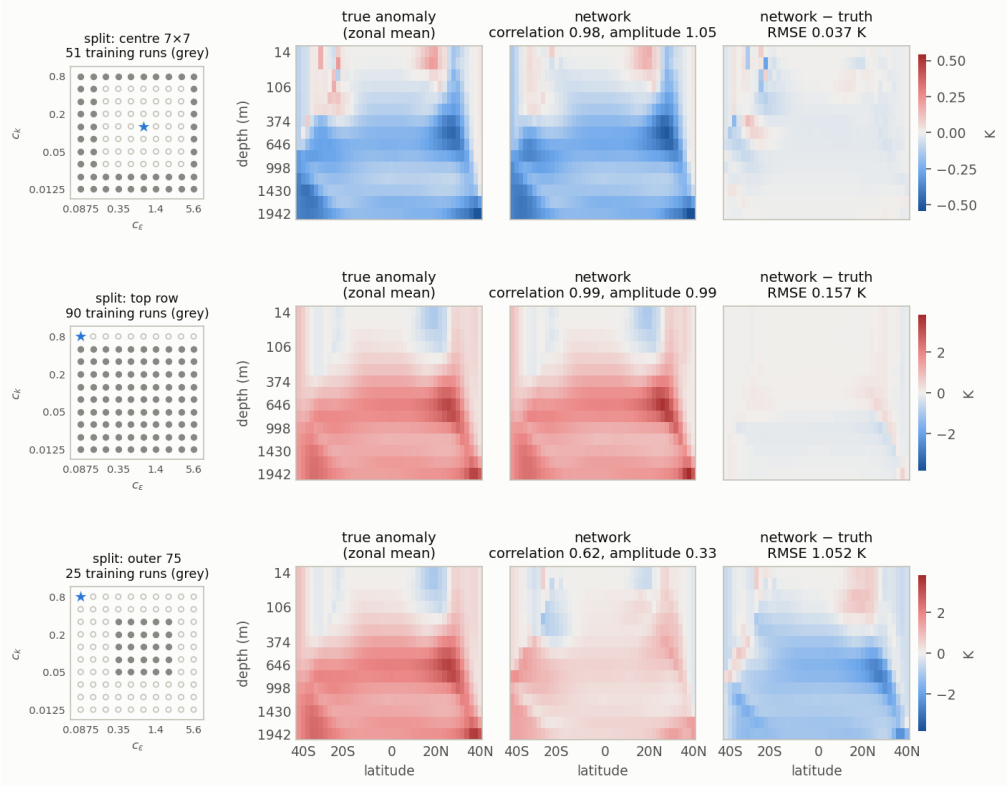


Figure 3: Zonal-mean anomaly of one held-out run relative to the mean of the training runs: truth, network and their difference. Top: centre of the held-out 7×7 block. Middle: warm corner, with the top row held out. Bottom: warm corner, with the 75 outer runs held out. Left: the split (grey: training; star: the run shown). Amplitude is the regression slope of the predicted on the true anomaly.

4.4 Restart from an emulated state

To test the emulated states as initial conditions, the model was restarted from them at eight held-out points, the corners of the outer-51 and of the 7×7 held-out sets, and run for ten years. Each restart starts from the predicted temperature with the model at rest, and is compared with the continuation of the true run past year 100. Distances are volume-weighted RMSEs of temperature.

At seven of the eight points the network's state lies 0.006–0.079 K from the true one, as close as the original run was after 92–99 of its 100 years (Figure 4). At the warm corner it lies 0.70 K away and is worth 56 years. The restart does not correct what the state gets wrong: after ten years between 0.5 and 1.9 times the initial error remains. Starting at rest has a cost of its own, 0.02–0.03 K within a year even from the true temperature, because the flow and the turbulent energies must be rebuilt. Away from the warm corner this cost is comparable to the error of the best emulators, so that after ten years the network, the GP on s and the nearest run are hard to tell apart: each is closest to the continuation at two or three of the eight points. No restart diverged.

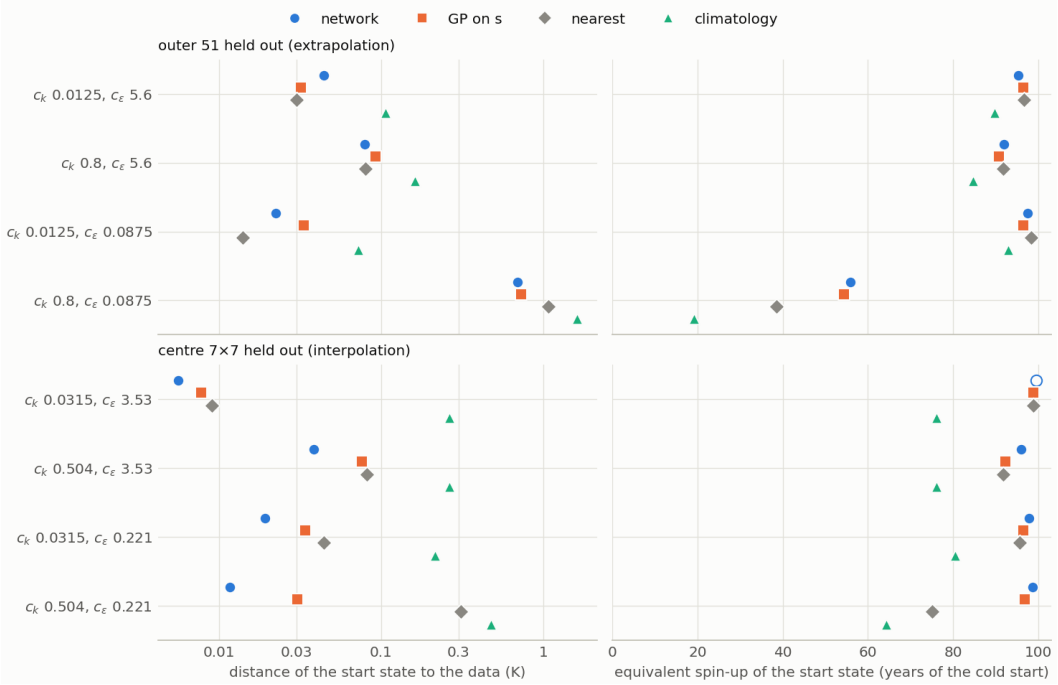


Figure 4: Start states of the restart experiment at eight held-out points. Left: distance to the true year-100 state. Right: equivalent spin-up, the year at which the original run first came that close to its year-100 state (hollow: 99 years or more).

5 Conclusions

The year-100 temperature of this model is predictable from the two mixing constants to a few hundredths of a kelvin inside the sampled range, from 51 to 99 training runs. Most of that skill comes from a one-dimensional structure that the closure itself explains: emulators built on s are the best, and the neural network improves on a Gaussian process on s by a margin that is consistent but not yet established. Beyond the sampled range every method fails together, because the predicted states cannot leave the span of the training states; more runs along the warm edge, or a decoder not confined to that span, are the remedies. As an initial condition an emulated state replaces most of the spin-up, but its error persists, and a restart from temperature alone adds an error comparable to that of the emulator. Predicting the flow and the turbulent energies along with temperature is the next step for that use.

A Hyper-parameters and their effect

The search covers two activations (ReLU, tanh) and five penalties ($\alpha = 10^{-4}$ to 1). Each configuration is scored by five-fold cross-validation on the training runs, repeated three times. The one-standard-error rule takes the most regularised configuration within one standard error of the best; the standard error is that of the best configuration's per-run errors. It selects ReLU with $\alpha = 10^{-2}$ on every split except outer 75 (10^{-1}) and four runs (1, Appendix B). The minimum of the same scores selects $\alpha = 10^{-3}$ or 10^{-4} (10^{-2} on outer 75). The rule was adopted after a first review had examined test errors, so the margins obtained with it are not a blind test; Table 2 gives both. For leave-one-out the choice is made once on all runs.

Table 3 gives the test error of the configurations that cross-validation cannot separate. The choice of the rule has the lowest test error among them on five of seven splits, and is unchanged when the cross-validation uses other partitions.

	one-standard-error rule (reported)			minimum CV RMSE				
split	difference [95 % interval]	p	Holm p	choice	network	difference	p	Holm p
scattered	-0.0025 [-0.0053, -0.0005]	0.037	0.074	10^{-3}	0.016	-0.0030	0.049	0.195
band (three rows)	-0.0075 [-0.0132, -0.0020]	0.015	0.044	10^{-3}	0.037	-0.0083	0.011	0.068
centre 5×5	-0.0045 [-0.0071, -0.0019]	0.003	0.013	10^{-4}	0.026	-0.0032	0.022	0.110
centre 7×7	-0.0058 [-0.0096, -0.0021]	0.003	0.013	10^{-3}	0.035	$+0.0010$	0.745	1.000
leave-one-out	-0.0046 [-0.0086, -0.0015]	< 0.001	< 0.001	10^{-3}	0.023	-0.0052	< 0.001	< 0.001
top row	-0.0297 [-0.0768, $+0.0001$]	0.193	0.193	10^{-4}	0.058	-0.0204	1.000	1.000
outer 51	-0.0076 [-0.0123, -0.0032]	0.002	0.013	10^{-3}	0.067	-0.0072	0.009	0.066
outer 75	-0.0111 [-0.0170, -0.0059]	< 0.001	0.004	10^{-2}	0.122	-0.0032	0.594	1.000

Table 2: Network minus GP on s, per held-out run (K; negative: network better): mean, 95 % bootstrap interval, Wilcoxon signed-rank p and Holm-corrected p over the eight splits. Left: the reported network. Right: the network chosen by the minimum cross-validation error (its α and mean RMSE). The tests treat runs as independent and are therefore optimistic.

	configurations within one standard error					choice, other partitions	
split	number	test RMSE	below GP on s	rank of the choice	all ten	one SE	minimum
scattered	5	0.016–0.017	5	3	0.016–0.034	same	same
band (three rows)	6	0.037–0.040	6	4	0.037–0.072	same	10^{-2}
centre 5×5	5	0.025–0.030	3	1	0.025–0.046	same	10^{-3}
centre 7×7	6	0.029–0.045	1	1	0.029–0.060	same	10^{-4}
top row	5	0.049–0.058	5	1	0.049–0.146	same	10^{-3}
outer 51	5	0.067–0.077	3	1	0.067–0.121	same	same
outer 75	6	0.114–0.144	3	1	0.114–0.165	same	same
four runs	6	0.117–0.187	2	5	0.098–0.187	–	–

Table 3: Configurations within one standard error of the best cross-validation score: their number, the range of their test RMSE (K), how many beat the GP on s, and the rank of the rule's choice among them. All ten: range over the whole search. Right: the choice of each rule with other cross-validation partitions.

B Four training runs

Training on four runs, one step in from each corner of the grid, leaves 96 runs to predict from three principal directions. Five-fold cross-validation is then leave-one-out, its standard error (0.15 K) covers six configurations, and both rules select the strongest penalty ($\alpha = 1$), which flattens the response. The network then scores 0.147 K (0.134–0.169 over the sets of initialisations), against 0.120 K for the GP on s, 0.146 K for the spline and 0.181 K for the nearest run. Over the ten configurations it ranges from 0.10 to 0.19 K: the outcome is set by a choice that four runs cannot inform.

C Reproducibility

Every fit runs with one BLAS thread, which makes a refit bit-identical (scikit-learn 1.7.2, NumPy 2.2.5, OpenBLAS 0.3.30). With eight threads L-BFGS stops elsewhere and a split mean moves by up to 0.0003 K. The evaluation takes 46 minutes on eight cores and the sensitivity analysis 28. Tables are generated from the results by `det/tables.py` and figures by `det/figures.py`; Figure 4 comes from the companion repository. A 100-year run of the model takes 12 minutes on a laptop and reproduces a stored run to 7×10^{-8} K at year 100.

References

[1] B. Blanke and P. Delécluse (1993). Variability of the tropical Atlantic Ocean simulated by a general circulation model with two different mixed-layer physics. *J. Phys. Oceanogr.* 23, 1363–1388.

[2] P. G. Constantine, E. Dow and Q. Wang (2014). Active subspace methods in theory and practice: applications to kriging surfaces. *SIAM J. Sci. Comput.* 36, A1500–A1524.

[3] C. Eden and R. J. Greatbatch (2008). Towards a mesoscale eddy closure. *Ocean Modelling* 20, 223–239.

[4] P. Gaspar, Y. Grégoris and J.-M. Lefevre (1990). A simple eddy kinetic energy model for simulations of the oceanic vertical mixing. *J. Geophys. Res.* 95, 16179–16193.

[5] P. R. Gent and J. C. McWilliams (1990). Isopycnal mixing in ocean circulation models. *J. Phys. Oceanogr.* 20, 150–155.

[6] D. Häfner et al. (2018). Veros v0.1 – a fast and versatile ocean simulator in pure Python. *Geosci. Model Dev.* 11, 3299–3312.

[7] D. Häfner, R. Nuterman and M. Jochum (2021). Fast, cheap, and turbulent – global ocean modeling with GPU acceleration in Python. *J. Adv. Model. Earth Syst.* 13, e2021MS002717.

[8] T. Hastie, R. Tibshirani and J. Friedman (2009). *The Elements of Statistical Learning*, 2nd ed., Springer, Section 7.10.

[9] D. Higdon, J. Gattiker, B. Williams and M. Rightley (2008). Computer model calibration using high-dimensional output. *J. Amer. Statist. Assoc.* 103, 570–583.

[10] S. Holm (1979). A simple sequentially rejective multiple test procedure. *Scand. J. Statist.* 6, 65–70.

[11] M. A. Mrozowska et al. (2025). Bayesian optimization with GPU acceleration for ocean models. *J. Geophys. Res. Mach. Learn. Comput.* 2, e2024JH000517.

[12] F. Pedregosa et al. (2011). Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.

[13] C. E. Rasmussen and C. K. I. Williams (2006). *Gaussian Processes for Machine Learning*. MIT Press.

[14] J. M. Salter, D. B. Williamson, J. Scinocca and V. Kharin (2019). Uncertainty quantification for computer models with spatial output using calibration-optimal bases. *J. Amer. Statist. Assoc.* 114, 1800–1814.